A candidate sharp multipartite-entanglement bound for CZX symmetry

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We give an elementary candidate proof of a sharp tripartite convex-roof entanglement bound for the anomalous CZX symmetry on a periodic qubit chain. For three nonempty connected arcs, the minimum over states in a single symmetry sector is $\frac{3}{2}h((2+\sqrt{2})/4)$, confirming a numerical conjecture of Lessa, Cheng, and Wang. Their construction already supplies candidate saturating bases; our lower-bound argument reduces each state to three encoded qubits and uses anticommuting observables. We give explicit saturation for a weighted extension and classify arbitrary three-party partitions by the parity of their inter-party edges. A sharp expectation-value witness yields an exact separability threshold for a globally depolarized charge sector. The results concern this particular lattice symmetry; specialist verification and research priority remain open.

Anomalous symmetries constrain the entanglement structure of quantum matter. The CZX action is a non-onsite boundary symmetry of a bosonic symmetry-protected phase [2, 3]. Lessa, Cheng, and Wang [1] proved tripartite nonseparability for strongly symmetric states and a separate long-range preparation obstruction. In Appendix A of Ref. [1], they conjectured a universal-in-size numerical minimum of a tripartite convex-roof measure for this action and gave complete bases of candidate minimizers. We address that quantitative conjecture. The anticommuting-observable and entropy-concavity techniques used below have well-established precedents [5, 6].

Consider $L \geq 3$ qubits on a ring, with $\mathrm{CZ}_{j,j+1} = \mathrm{diag}(1,1,1,-1)$ and

$$U = \Big(\prod_{j=1}^{L} X_j\Big)\Big(\prod_{j=1}^{L} \mathrm{CZ}_{j,j+1}\Big), \quad \widehat{U} = i^{L \bmod 2} U, \quad \widehat{U}^2 = \mathbb{I}. \tag{1}$$

The odd-L phase merely makes the charge eigenvalues ± 1; the state and its entanglement are unchanged. Fix three disjoint nonempty subsets A, B, C whose union is the ring, and take all logarithms to be natural. For $w_R \geq 0$, define for pure states and, by convex roof, for mixed states

$$\begin{aligned} \mathcal{F}_w(\psi) &= \sum_{R=A,B,C} w_R S(\rho_R), \\ \mathcal{F}_w(\rho) &= \inf_{\rho=\sum_j p_j |\psi_j\rangle\langle\psi_j|} \sum_j p_j \mathcal{F}_w(\psi_j). \end{aligned} \tag{2}$$

Here $\rho_R = \mathrm{Tr}_{R^c} |\psi\rangle\langle\psi|$. The source's E_3 [1, 4] is $\mathcal{F}_w$ with $w_A = w_B = w_C = 1/2$. Strong symmetry means $\widehat{U}\rho = \lambda\rho$ for $\lambda = \pm 1$; commutation with $\widehat{U}$ alone is not sufficient.

Let e_{RS} be the number of ring edges between R and S, modulo 2. The boundary of each subset of a cycle has even size, so

$$e_{AB} = e_{BC} = e_{CA} =: e \in \{0, 1\}. \tag{3}$$

This parity is the mod-two winding of the party-label sequence around the triangle $A \to B \to C \to A$. Write $W = \sum_R w_R$, $w_{\max} = \max_R w_R$, and

$$\begin{aligned} h(x) &= -x \log x - (1-x)\log(1-x), \\ p &= \frac{2+\sqrt{2}}{4}, \quad B_w = \min\{(W - w_{\max})\log 2, W h(p)\}. \end{aligned} \tag{4}$$

Our candidate result is

$$\begin{aligned} \mathcal{F}_w(\rho) &\geq e B_w, \qquad \rho_\lambda = \frac{\mathbb{I} + \lambda \widehat{U}}{2^L}, \\ \mathcal{F}_w(\rho_\lambda) &= e B_w. \end{aligned} \tag{5}$$

The first inequality assumes strong symmetry. When $e = 1$, each value in the minimum is attained by a pure state in either sector. For $e = 0$, each sector projector has a complete product eigenbasis across $A|B|C$. This does not make every state in the sector separable.

Three connected arcs satisfy $e = 1$. In that case Eq. (5) gives

$$E_3(\rho) \geq E_3(\rho_\lambda) = \frac{3}{2} h(p) = 0.9013140550\ldots \log 2. \tag{6}$$

The inequality and equality supply the lower bound and projector value conjectured in Ref. [1]; its saturating bases and qualitative nonseparability theorem are prior results. The $e = 1$ statement also applies to disconnected parties with the same edge parity. For $e = 0$, the absence of a positive lower bound is consistent with the known disconnected-partition caveat [1]. Equation (6) concerns separability across the specified three parties, not genuine multipartite entanglement: the projector can be separable across each bipartition.

Local block reduction.— In the computational basis,

$$U|b_1 \cdots b_L\rangle = (-1)^{\sum_j b_j b_{j+1}} |\bar{b}_1 \cdots \bar{b}_L\rangle. \tag{7}$$

For every complement pair $\{\alpha_R, \bar{\alpha}_R\}$ in party R, the projector $P_\alpha^R = |\alpha_R\rangle\langle\alpha_R| + |\bar{\alpha}_R\rangle\langle\bar{\alpha}_R|$ commutes with $\widehat{U}$. The products of these party-local projectors form 2^{L-3} invariant eight-dimensional blocks, each described by three encoded qubits. An eigenstate remains in its charge sector after any nonzero outcome.

These measurements cannot increase the average $\mathcal{F}_w$ of a pure state. For example, when measuring A, the average reduced states of B, C, and BC are unchanged. Entropy concavity bounds the average S_B, S_C, and $S_{BC} = S_A$ by their initial values. Repeat for the other parties. The operation is local relative to $A|B|C$, but may be spatially nonlocal within a party; it does not supply a finite-depth preparation theorem.

Within a block let x, y, z label a representative bitstring or its complement in each party. Equation (7) becomes

$$q = q_0 + \ell_A x + \ell_B y + \ell_C z + e(xy + yz + zx) \pmod 2. \quad (8)$$

Internal edges contribute only constant and linear terms. Since $q(x,y,z) + q(1-x,1-y,1-z) = L \pmod 2$, one has $\ell_A + \ell_B + \ell_C = e + L \pmod 2$. If $e = 0$, the block operator $\widehat{U}$ is a sign times a product of single-qubit X or Y operators: $XZ = -iY$ and the stated parity make the overall phase real. Each of its charge sectors therefore has four orthonormal product eigenvectors. These bases, assembled over all blocks, prove the $e = 0$ claims in Eq. (5).

For $e = 1$ and even L, the linear vector ℓ has odd parity. Conjugating by an encoded X_R flips the other two linear coefficients; such flips can bring ℓ to $(1,0,0)$, up to a scalar sign. For odd L first conjugate $\widehat{U}$ by $S_A = \mathrm{diag}(1, i)$, changing the phase polynomial to $q + x + 1$; the same reduction applies. Thus each block is locally unitarily equivalent, up to sign, to either charge sector of

$$V = iX_A X_B X_C \ \mathrm{CZ}_{AB}\, \mathrm{CZ}_{BC}\, \mathrm{CZ}_{CA}, \quad V^2 = \mathbb{I}. \quad (9)$$

In particular, for the canonical operator $U_{100}|xyz\rangle = (-1)^{x+xy+yz+zx}|\bar{x}\bar{y}\bar{z}\rangle$, $S_A V S_A^\dagger = -U_{100}$.

Anticommutation and the lower bound.— Let $|\psi\rangle$ be a V eigenstate. Since $VZ_RV^\dagger = -Z_R$, each reduced Bloch vector lies in the XY plane. Let r_R be its length and choose a unit XY Pauli operator M_R with expectation r_R (arbitrarily if $r_R = 0$). For $M_A = aX_A + bY_A$,

$$VM_AV^\dagger = (aX_A - bY_A)Z_BZ_C. \quad (10)$$

This Hermitian involution anticommutes with M_B; the eigenstate condition makes its expectation r_A. Squaring a real linear combination of these two observables gives

$$r_A^2 + r_B^2 \le 1, \quad r_B^2 + r_C^2 \le 1, \quad r_C^2 + r_A^2 \le 1. \quad (11)$$

This is the familiar anticommuting-observable method [5]; here the three inequalities arise from different pairs of observables.

Set $x_R = r_R^2$. The single-qubit entropy is the concave function

$$f(x) = h\left(\frac{1+\sqrt{x}}{2}\right) = \log 2 - \sum_{n\ge1} \frac{x^n}{2n(2n-1)}. \quad (12)$$

The polytope defined by Eq. (11) has vertices 0, the three coordinate unit vectors e_R, and $(1/2, 1/2, 1/2)$. Indeed, if $m = \min_R x_R$, the coefficients in

$$x = \big[1 - \sum_R x_R + m\big]0 + \sum_R (x_R - m)e_R + 2m(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}) \quad (13)$$

are nonnegative and sum to one. Concavity of $\sum_R w_R f(x_R)$ gives the minimum of the vertex values $W \log 2$, $(W - w_R)\log 2$, and $Wh(p)$, which is B_w. The local measurements extend this bound to every pure charge eigenstate. For a strongly symmetric mixed state, every vector in every pure ensemble lies in its charge sector by positivity; taking the convex roof proves the first line of Eq. (5).

Attainment.— In the $+1$ sector of U_{100}, the vector

$$v_* = (1, 1, 1, i, -i, 1, 1, 1)/\sqrt{8} \quad (14)$$

is given in the order $000, 001, \ldots, 111$. Each marginal has eigenvalues $p, 1-p$. Acting with $\mathbb{I}, Z_AZ_B, Z_AZ_C, Z_BZ_C$ produces four orthonormal charge eigenvectors with the same marginal entropies. Orthogonality follows from the uniform computational-basis probabilities of v_*.

For the other branch, the following unnormalized vectors form an orthonormal basis after division by 2:

$$\begin{aligned} &(1,1,0,0,0,0,1,1), \quad (1,-1,0,0,0,0,-1,1), \\ &(0,0,1,i,-i,1,0,0), \quad (0,0,1,-i,i,1,0,0). \end{aligned} \quad (15)$$

Each lies in the $U_{100} = +1$ sector, has C pure, and has maximally mixed A and B. Permutation symmetry in the V representation and the local conjugation above give the other pure-party choices. A local Z_R changes the charge without changing entropies. Transporting either four-vector basis into each physical block yields a complete orthonormal basis of each charge sector. The uniform mixture is ρ_λ; using the basis realizing the smaller branch proves the second line of Eq. (5). For equal weights $Wh(p) < \log 2$, giving Eq. (6).

A sharp separability witness.— For any fully separable state across an $e = 1$ partition,

$$|\,\mathrm{Tr}(\rho\widehat{U})| \le \frac{1}{\sqrt{2}}. \quad (16)$$

The three-qubit version follows from the established Gühne–Han–Kye separability criterion [9, 10]; the step here lifts it to the original ring and its partitions. In the canonical block, $V = (YXX + XYX + XXY + YYY)/2$. For a product state write its XY Bloch vectors as $r_R(\cos\theta_R, \sin\theta_R)$ with $r_R \le 1$. Optimizing over θ_A yields

$$|\langle V\rangle| \le \frac{1}{2}\sqrt{\sin^2(\theta_B + \theta_C) + \cos^2(\theta_B - \theta_C)} \le \frac{1}{\sqrt{2}}. \quad (17)$$

Equality holds when all three angles are $\pi/4$. Party-local block measurements preserve product states and the

expectation of $\widehat{U}$; convexity and the block equivalences prove Eq. (16) and its sharpness. This detects nonseparability across all three specified parties, not genuine tripartite entanglement.

The same observable gives a quantitative bound for arbitrary mixed states:

$$E_3(\rho) \geq \tfrac{1}{4}\left[|\operatorname{Tr}(\rho\widehat{U})| - 1/\sqrt{2}\right]_+^2. \tag{18}$$

For a pure state ψ choose the product of its marginals, $\sigma = \rho_A \otimes \rho_B \otimes \rho_C$. The witness and quantum Pinsker's inequality give $|\langle\widehat{U}\rangle_\psi| - 1/\sqrt{2} \leq \|\psi - \sigma\|_1 \leq \sqrt{2D(\psi\|\sigma)} = 2\sqrt{E_3(\psi)}$. The squared positive-part function of the expectation is convex, so Jensen's inequality and the convex roof prove Eq. (18). Its use requires estimating the global symmetry expectation, generally via a coherent global measurement or circuit.

More strongly, the globally depolarized sector projector $\rho(v) = (\mathbb{I} + v\widehat{U})/2^L$, $|v| \leq 1$, is fully separable across an $e = 1$ partition *if and only if*

$$|v| \leq 1/\sqrt{2}. \tag{19}$$

To prove sufficiency at $v = 1/\sqrt{2}$, in each V block average the eight product states whose local Bloch vectors are $(s_R, r s_R, 0)/\sqrt{2}$, with $r = \pm 1$, $s_R = \pm 1$ and $s_A s_B s_C = r$. Their average is $(\mathbb{I} + V/\sqrt{2})/8$: all one- and two-party correlators vanish and each of YXX, XYX, XXY, YYY has expectation $1/(2\sqrt{2})$. Local Z gives the negative endpoint; mixing the endpoints gives every intermediate v. Transporting these states to each orbit block proves Eq. (19) on the full ring. For $e = 0$, the product eigenbasis makes $\rho(v)$ separable for every $|v| \leq 1$. Reference [1] already studied the sector projector by a permutation criterion. Local postselection on a complement block preserves separability; applying its index permutation to that normalized block at general v gives trace norm $1/2 + |v|/\sqrt{2}$, implying the necessary side of Eq. (19). Moreover, Chen *et al.* already studied the same three-qubit GHZ-diagonal family and its $1/\sqrt{2}$ boundary [7], and Han and Kye established a general GHZ-diagonal separability criterion [8]. Our calculation makes the ring-block mapping, product ensemble, and arbitrary-partition scope explicit; it does not claim the three-qubit boundary is new. The parameter v describes global depolarization. For independent Z flips of probability q at every site, $\rho(1)$ instead becomes $\rho((1-2q)^L)$, since each Z_j anticommutes with $\widehat{U}$. On an odd-parity cut and for $0 \leq q \leq 1/2$, Eq. (19) gives the exact separability threshold $q_c = [1 - 2^{-1/(2L)}]/2 \sim (\log 2)/(4L)$. This is a statement about the chosen partition, not a finite-depth preparation threshold.

The relative entropy to the fully separable set for the same family is also explicit. With $a = 1/\sqrt{2}$, $P = (1 + |v|)/2$, and $Q = (1 + a)/2$, it is zero for $|v| \leq a$ and otherwise equals

$$E_R(\rho(v)) = P \log \frac{P}{Q} + (1 - P) \log \frac{1 - P}{1 - Q}. \tag{20}$$

Measuring $\widehat{U}$ and using Eq. (16) gives the lower bound by data processing; the separable $\rho(\operatorname{sgn}(v)a)$ attains it. Reference [7] gives the corresponding three-qubit formula. Equation (20) is its ring lift and is a different measure from E_3.

Sector leakage.— For an arbitrary state on an $e = 1$ partition, put $P_\pm = (\mathbb{I} \pm \widehat{U})/2$ and $\epsilon = \min\{\operatorname{Tr}(P_+\rho), \operatorname{Tr}(P_-\rho)\}$. Then

$$\mathcal{F}_w(\rho) \geq \left[B_w - W h(\min\{\sqrt{\epsilon}, 1/2\})\right]_+. \tag{21}$$

For each postmeasurement pure block with leakage $\delta < 1$, normalize its projection to the chosen sector. Its trace distance from the original block state is $\sqrt{\delta}$; partial trace and qubit entropy continuity [11] bound each change of entropy by $h(\min\{\sqrt{\delta}, 1/2\})$. The case $\delta = 1$ is trivial. The capped function is concave and nondecreasing, and the projectors commute with the sector measurement. Jensen's inequality over blocks and then over any pure ensemble proves Eq. (21). In particular $E_3 > 0$ whenever $\epsilon < (3 - 2\sqrt{2})/8 \simeq 0.02145$. The sharper witness alone detects nonseparability for $\epsilon < (1 - 1/\sqrt{2})/2 \simeq 0.14645$; it does not give Eq. (21)'s quantitative entropy lower bound. These are bounds in total charge leakage, not in a local error rate.

More than three parties.— For any partition into $m \geq 3$ nonempty parties, let e_{ij} again count crossing edges modulo two. If all $e_{ij} = 0$, the local complement-block action factorizes, and each sector projector has a complete product eigenbasis. If some $e_{ij} = 1$, group parties i, j, and the remaining parties into three sets. Each set has even cut degree on the ring, so the three grouped crossing parities are odd. Equation (16) then applies to every m-party product mixture, although its bound need not be sharp when $m > 3$. Thus strongly symmetric fully m-separable states exist exactly when all crossing parities vanish. This is an existence and witness statement; no m-party weighted convex-roof formula is asserted.

Scope and status.— These numerical constants apply to the specified CZX representative and partition; they are not asserted to be universal anomaly invariants. The maximally mixed $\mathbb{I}/2^L$ is weakly symmetric and fully separable, demonstrating the need for charge-sector information. Exact finite-size checks of the block signs and saturation bases are supporting diagnostics, not a substitute for the all-size argument. This draft and its priority require independent specialist review. OpenAI Codex (including GPT-6.1 Terra subagents) assisted the derivation, literature search, analytic and finite-size checks, and drafting, under user direction; independent human verification and author identification are pending.

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